Wireline Log QC and Basic Formation Evaluation

Well 15/9-19 SR - Volve field area, Norwegian North Sea (real data, Equinor Volve open data set - see DATA_SOURCE.md for provenance and licence).
Analysis interval 3568-4630 m MD where GR, density, neutron and resistivity all exist; 6966 samples at 0.1524 m. Target zone: HUGIN FM.

1. Summary

Data acceptance: **ACCEPT WITH FLAGS** - 2 of 7 checks need review. 422 QC events were logged; flagged samples were excluded from interpretation (short gaps up to ~1 m interpolated, longer gaps left empty). Raw data is never modified.

2. Data inventory (full file)

Curve	LAS	Unit	From (m)	To (m)	Samples	Min	Median	Max
GR	GR	GAPI	102	4635	28117	2.77	75.00	304.33
RHOB	DEN	G/CC	3550	4630	7084	1.94	2.51	3.00
NPHI	NEU	%	3550	4631	7096	0.02	0.16	1.46
RT	RDEP	OHMM	106	4637	12223	0.25	1.54	198.54
RM	RMED	OHMM	106	4637	12223	0.18	1.49	168.60
DT	AC	US/F	3550	4618	7007	1.03	74.66	181.81
CALI	CALI	IN	3550	4618	7007	6.00	9.59	20.33

GR runs the whole well; density, neutron and resistivity cover the lower part only, so the analysis is limited to the interval where all are present.

3. Data acceptance checks

Check	Status	Remarks
Required curves present (GR, density, neutron, resistivity)	Pass	All four main curves found
Regular depth sampling, no logging skips	Pass	nominal step 0.1524 m; 0 gap(s)
Null values below 5 % on main curves (in logged interval)	Pass	worst curve 0.0 % null
Out-of-range samples below 1 % of interval	Review	185 samples (2.66 %)
Washout below 10 % of interval	Pass	2.0 % of interval has enlarged hole
Valid data over HUGIN FM (target) above 90 %	Pass	100 % valid after QC
Tool calibration records and repeat section	Not available	Not included in the LAS file - request from logging company

4. QC summary by curve (analysis interval)

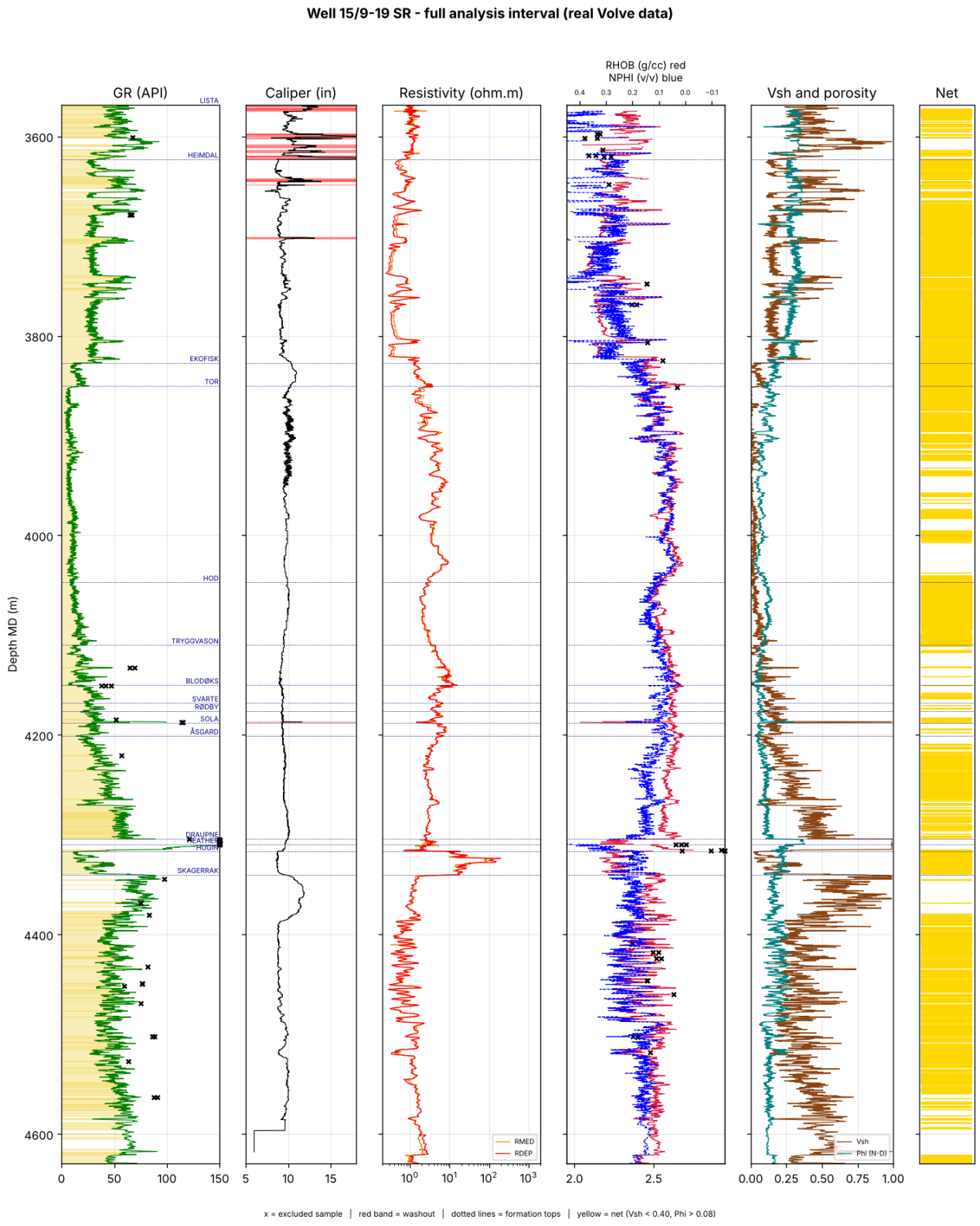
Curve	Null %	Out-of-range samples	Spike events	Washout %	Excluded %
GR	0.0	25	21	-	0.85
RHOB	0.0	1	26	-	2.6
NPHI	0.0	159	148	-	5.91
RT	0.0	0	0	-	0.0
RM	0.0	0	6	-	0.09
DT	1.11	15	45	-	2.37
CALI	1.11	5	72	2.05	2.61

Checks: nulls; physical ranges; narrow spikes (rolling median, 8 robust sigma, max 3 samples); caliper washout (> 1.5 in above local gauge) - density and neutron are excluded in washouts; depth skips. Full event list: output/qc_flags.csv.

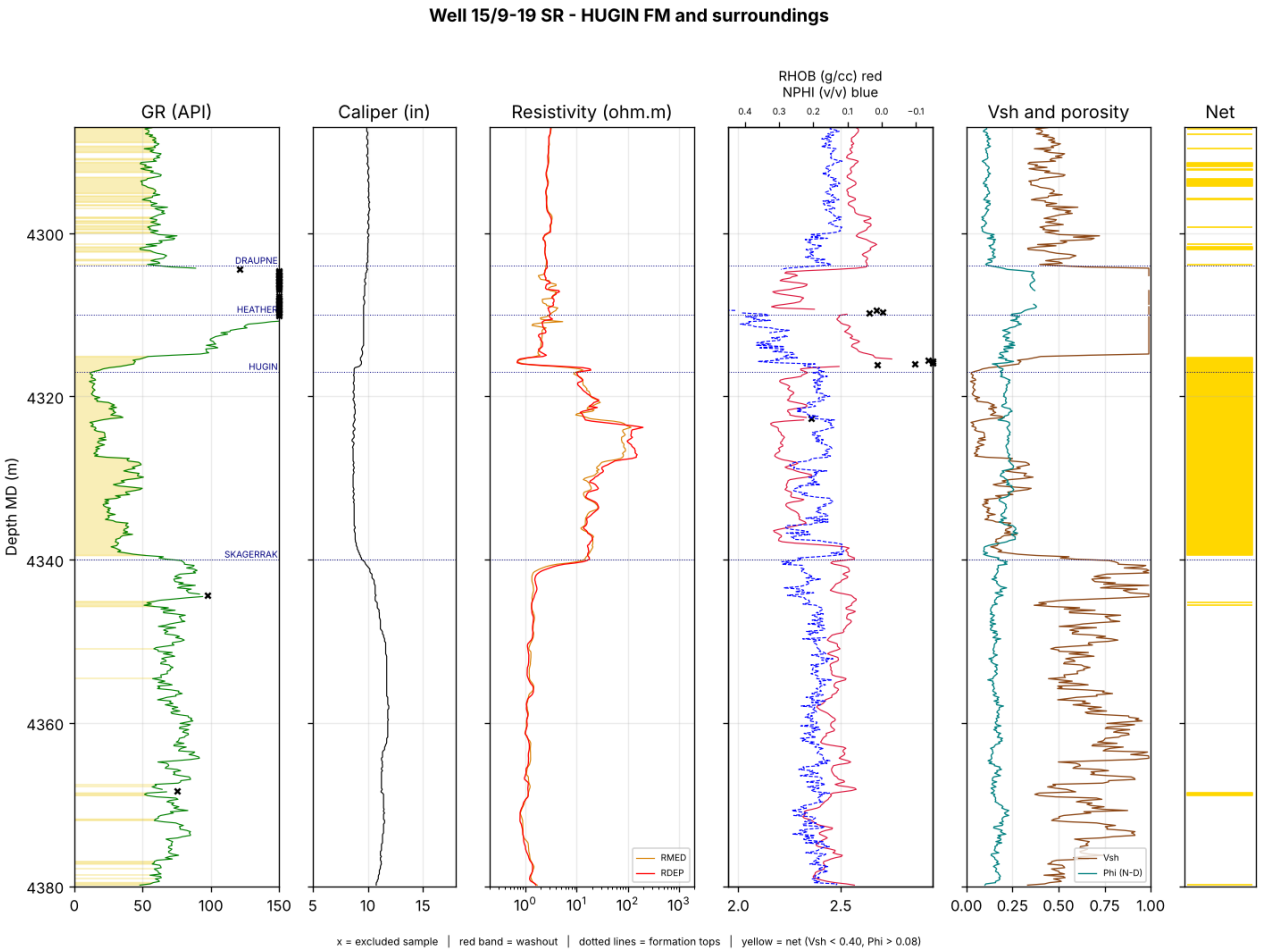
Ten largest QC events

Curve	Issue	Top (m)	Base (m)	Samples	Detail
CALI	Missing / null	4618.1	4629.7	77	77 null samples
DT	Missing / null	4618.1	4629.7	77	77 null samples
CALI	Washout / enlarged hole	3569.7	3572.6	20	caliper up to 13.5 in vs gauge 10.3 in
NPHI	Out of range	3618.8	3621.2	17	min 0.60 / max 1.06, valid 0.0-0.6
CALI	Washout / enlarged hole	3620.5	3622.9	17	caliper up to 20.3 in vs gauge 9.9 in
NPHI	Out of range	4306.3	4308.6	16	min 0.60 / max 0.83, valid 0.0-0.6
CALI	Washout / enlarged hole	3598.2	3600.5	16	caliper up to 20.3 in vs gauge 10.4 in
CALI	Washout / enlarged hole	3608.7	3610.7	14	caliper up to 13.3 in vs gauge 10.1 in
GR	Out of range	4304.9	4306.9	14	min 254.22 / max 295.72, valid 0-250
CALI	Washout / enlarged hole	3614.2	3615.9	12	caliper up to 14.3 in vs gauge 10.0 in

5. Log display - full analysis interval



6. Log display - HUGIN FM and surroundings



Method and assumptions. Vsh: Larionov (older rocks) from a gamma-ray index with baselines at the 5th/99th percentile of cleaned GR (7 / 89 API) for the whole interval. Porosity: density porosity (matrix 2.65 g/cc sand, 2.71 g/cc for Ekofisk/Tor/Hod chalk, fluid 1.0 g/cc) averaged with neutron porosity by root-mean-square; no shale, gas or lithology corrections. Net rock = Vsh < 0.40 and porosity > 0.08. Net is a rock-quality flag, **not** net pay: no water saturation is computed because no reliable Rw is available.

7. Formation summary

Formation	Top	Base	Thk	Valid %	GR	Vsh	Phi	RT med	Net m	NTG
LISTA FM	3568.2	3623.0	54.8	74.0	56.0	0.44	0.31	1.1	19.2	0.48
HEIMDAL FM	3623.0	3827.0	204.0	99.0	37.0	0.23	0.288	0.4	173.6	0.86
EKOFISK FM	3827.0	3850.0	23.0	100.0	16.0	0.05	0.159	1.5	23.0	1.0
TOR FM	3850.0	4047.0	197.0	100.0	9.0	0.01	0.082	3.7	88.8	0.45
HOD FM	4047.0	4110.0	63.0	100.0	15.0	0.05	0.113	2.3	60.8	0.96
TRYGGVASON FM	4110.0	4150.0	40.0	100.0	18.0	0.07	0.059	6.9	4.1	0.1
BLODØKS FM	4150.0	4168.0	18.0	100.0	28.0	0.15	0.062	4.8	3.8	0.21
SVARTE FM	4168.0	4176.0	8.0	100.0	29.0	0.15	0.067	4.6	0.9	0.11
RØDBY FM	4176.0	4188.0	12.0	100.0	36.0	0.23	0.088	4.5	2.1	0.18
SOLA FM	4188.0	4201.0	13.0	100.0	30.0	0.16	0.061	6.4	1.4	0.1
ÅSGARD FM	4201.0	4304.0	103.0	100.0	44.0	0.3	0.099	3.0	53.2	0.52
DRAUPNE FM	4304.0	4310.0	6.0	44.0	174.0	0.97	0.313	2.8	0.0	0.0
HEATHER FM	4310.0	4317.0	7.0	98.0	93.0	0.76	0.223	2.2	1.8	0.27
HUGIN FM	4317.0	4340.0	23.0	100.0	28.0	0.15	0.213	19.8	22.6	0.98
SKAGERRAK FM	4340.0	4629.7	289.7	100.0	55.0	0.43	0.155	1.2	141.3	0.49

Net rock intervals in HUGIN FM

Top	Base	Thk (m)	Vsh	Phi	RT med	Indicator
4317.1	4339.6	22.6	0.15	0.214	20.2	RT > 10 ohm.m

A resistivity indicator means RDEP above 10 ohm.m in clean porous rock; it needs a saturation calculation and fluid data to be confirmed.

8. Pre-job readiness checklist (illustrative template)

Category	Item	Status	Remarks
Tool & Equipment	Tool string identified and serial numbers recorded	Done	GR / Resistivity / Density-Neutron string
Tool & Equipment	Surface and downhole function tests completed	Done	All sensors responding
Tool & Equipment	Gamma ray tool calibration certificate valid	Done	Calibration within validity period
Tool & Equipment	Density-neutron tool calibration verified	Pending	Calibration record not yet sighted
Tool & Equipment	Cable / wireline unit inspection completed	Done	Head tension and depth encoder checked
Tool & Equipment	Spares and consumables inventory checked	Done	O-rings and cable heads available
Documentation	Job program and well data sheet received	Done	
Documentation	Depth reference (KB/RT elevation) confirmed	Done	
Documentation	Previous-run log files / offset logs reviewed	Not Done	Not requested from client
Safety (HSE)	Pre-job safety meeting (toolbox talk) held	Done	
Safety (HSE)	Job safety analysis / risk assessment signed	Done	
Safety (HSE)	Radioactive source custody and handling log verified	Pending	Awaiting source handler sign-off
Safety (HSE)	PPE and barricading in place at rig floor	Done	
Safety (HSE)	Emergency response and stop-work authority briefed	Done	

Result: **NOT READY - 3 of 14 checks open**. These statuses are a worked example for a wireline job; they are not records from the real Volve job.

9. HSE risk register (generic wireline hazards)

Hazard	Activity	L	S	Score	Rating	Controls	Owner
Dropped object / pinch point	Rig-up and rig-down at rig floor	3	4	12	High	Exclusion zone; hard hats; tag lines; communication with driller	Field Engineer
Radioactive source exposure	Density/neutron source handling	2	5	10	Medium	Source custody log; barricading; shielded transport container; trained handler only	Wireline Specialist
Cable or tool stuck in hole	Logging run through tight hole	2	4	8	Medium	Monitor head tension; weak-point rating checked; stuck-tool procedure briefed	Field Engineer
High-pressure or hot surface contact	Lubricator and pressure equipment	2	4	8	Medium	Pressure test records; PPE; flagged hot zones	Wireline Specialist
Slips / trips / manual handling	Equipment loading and rig-up	4	2	8	Medium	Housekeeping; team lifting; clear walkways	All crew
Well control event (kick)	Open-hole logging	1	5	5	Low	Confirm hole fill-up with driller; BOP readiness; stop-work authority	Company Man

Score = Likelihood x Severity: 1-5 Low, 6-11 Medium, 12+ High.

10. Limitations and unresolved issues

• Calibration records, repeat section and the logging contractor's own QC are not in the dataset; the data cannot be formally accepted without them.

• Neutron porosity is assumed to be a limestone/sandstone-scaled v/v value; the LAS header does not say which matrix the tool was calibrated to.

• No water saturation, core calibration or fluid data - net rock is not net pay.

• Washout flags depend on a rolling-median gauge estimate, not the true bit size.

• Thin beds below the sampling and tool resolution are not resolved.